

I considered this scenario:

I am assuming that the dataset is being **prepared for a CycleGAN-based model** that can **generate synthetic CT scans from MRI images**. The clinical motivation behind this is MRI-only radiotherapy planning, where CT scans are usually required to calculate radiation dose. It could reduce the need for CT scans and therefore reduce unnecessary radiation exposure for patients.

Reproducibility Measures

1. All preprocessing parameters (such as image size, normalization ranges, and split ratios) are stored in a central configuration dictionary, making it easy to review or modify the setup in one place.
2. A **fixed random seed (42)** is used when splitting the dataset so that the train/validation/test split remains consistent across runs.
3. Each run of the pipeline **generates a JSON log file** that records the configuration used, timestamps, and details of how each volume was processed.

PreProcessing Step

1. The pipeline starts with file discovery and basic validation. **Each NIFTI file is checked to make sure it loads properly, is not empty, and does not contain invalid values such as NaNs.** Files corrupted or incomplete are skipped.
2. For CT scans, I applied **Hounsfield Unit (HU) windowing** using a brain-specific window (centre = 40, width = 80).
3. For MRI scans, I used **percentile-based clipping** (1st to 99th percentile) followed by normalization. More robust than min.max normalization
4. **Slice quality filter (less than 2% non-zero pixels).** These are usually empty slices from the top or bottom of the scan and do not contribute useful learning signal to the model.
5. Finally, the **dataset is split into training (70%), validation (15%), and testing (15%) sets**, ensuring that the split is done at the volume (patient) level rather than the slice level to avoid data leakage.

Data Quality Issue

- Corrupted or incomplete files were detected using file size checks and exception handling during loading. These files were skipped and logged for reference.
- All-zero volumes, which likely result from failed exports, were removed since they do not contain any useful information.
- NaN values were treated as critical errors. Any volume containing NaNs was excluded to prevent instability during training.
- Blank slices (mostly from the top and bottom of scans) were filtered out using a simple threshold based on foreground pixel percentage.
- 4D volumes (such as accidentally included fMRI data) were simplified by selecting a single time point.
- Inconsistent image sizes across different scans were handled by resizing all slices to a fixed resolution of 256×256 .

Key Risks for Clinical Deployment

1. Dataset Shift

Scans from different hospitals or machines can look quite different due to variations in scanners and protocols. A model trained on one dataset may not generalize well to another.

2. Lack of Ground Truth for Validation

Since the dataset is unpaired, there is no direct way to compare the generated CT scans with real CT scans from the same patient. This makes it difficult to measure how accurate the model actually is.